

ALATYR, Russian Federation

WMO: 276790

Lat: 54.817N

Lon: 46.583E

Elev: 180

StdP: 99.18

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.6	-23.3	-29.1	0.3	-26.5	-26.0	0.4	-23.1	11.0	-9.0	10.0	-4.7	2.9	270	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	30.8	19.7	28.8	19.0	27.1	18.4	21.0	28.1	20.1	26.9	19.3	25.5	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.7	17.7	13.0	22.8	16.9	12.3	22.5	61.5	28.0	58.5	26.9	55.5	25.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.5	7.4	-28.8	33.3	3.4	1.9	-31.2	34.7	-33.2	35.8	-35.1	36.9	-37.6	38.3
			-28.9	23.0	3.3	1.1	-31.3	23.9	-33.3	24.5	-35.2	25.2	-37.6	26.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-10.2	-9.7	-3.7	6.1	13.9	17.8	20.1	18.2	12.2	5.2	-2.2	-7.7
	DBStd	12.13	6.87	6.78	4.71	5.38	5.17	4.33	3.61	4.09	4.41	4.63	5.40	6.79
	HDD10.0	2869	627	551	426	142	21	2	0	0	26	160	367	548
	HDD18.3	5034	885	784	684	367	153	62	23	53	191	408	617	807
	CDD10.0	1072	0	0	0	24	143	234	314	255	91	11	0	0
	CDD18.3	196	0	0	0	0	17	45	78	49	6	0	0	0
	CDH23.3	1632	0	0	0	3	164	369	638	409	48	0	0	0
	CDH26.7	484	0	0	0	0	33	109	202	132	7	0	0	0
Wind	WSAvg	3.4	3.8	3.9	3.8	3.5	3.4	3.1	2.6	2.8	2.9	3.5	3.6	4.0
Precipitation	PrecAvg	471	31	24	27	28	35	59	54	54	51	40	37	33
	PrecMax	626	55	54	77	69	59	126	119	110	145	71	74	64
	PrecMin	290	10	5	3	3	8	11	4	11	8	16	10	9
	PrecStd	83	11	12	16	18	13	31	26	29	35	15	16	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	2.3	9.1	23.6	29.7	32.7	34.0	34.1	27.7	19.1	10.5	4.4
		MCWB	1.0	1.0	5.6	14.1	17.3	19.8	19.5	20.4	17.2	13.1	8.7	3.7
	2%	DB	1.0	1.4	5.3	20.5	27.2	29.6	30.8	29.9	24.4	15.9	7.9	2.5
		MCWB	0.3	0.5	2.8	12.4	16.8	18.9	19.9	19.8	16.3	11.1	6.7	1.7
	5%	DB	0.1	0.6	3.3	17.6	25.1	27.4	28.8	27.6	21.7	13.4	5.8	1.4
		MCWB	-0.4	0.0	1.4	10.9	15.9	18.3	19.7	18.7	15.2	10.4	5.0	0.8
	10%	DB	-1.0	-0.4	2.0	14.7	22.7	25.1	26.9	25.3	19.2	11.4	4.1	0.5
		MCWB	-1.5	-1.0	0.7	9.4	14.7	17.4	19.0	17.9	14.1	9.1	3.3	-0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.3	6.0	14.9	19.0	21.4	22.9	21.9	19.1	13.6	9.0	3.7
		MCDB	1.7	1.7	8.5	22.1	27.6	28.6	28.7	29.0	24.9	16.9	10.1	4.2
	2%	WB	0.5	0.7	3.2	13.1	17.5	20.2	21.4	20.5	17.0	12.0	6.9	1.8
		MCDB	0.8	1.2	4.9	19.3	25.4	27.1	27.7	28.0	22.6	15.2	7.7	2.3
	5%	WB	-0.4	0.0	1.7	11.4	16.5	19.3	20.5	19.5	15.6	10.5	5.1	0.9
		MCDB	0.1	0.6	2.8	16.7	23.9	25.7	26.8	26.2	21.1	13.0	5.8	1.3
	10%	WB	-1.5	-1.0	0.9	9.7	15.3	18.2	19.6	18.5	14.4	9.3	3.4	-0.1
		MCDB	-0.9	-0.3	1.8	14.3	21.8	23.9	25.7	23.9	18.6	11.2	4.0	0.4
Mean Daily Temperature Range	5% DB	MDBR	5.3	6.4	6.5	8.9	10.1	9.6	9.7	9.6	8.3	6.2	4.4	4.8
		MCDBR	4.3	4.5	7.0	13.2	13.0	12.3	12.4	12.7	12.6	9.9	5.8	3.5
		MCWBR	4.1	4.1	5.3	7.4	6.1	5.5	5.4	5.3	6.4	6.2	5.2	3.4
	5% WB	MCDBR	4.1	4.3	5.9	12.4	12.2	10.9	10.6	11.3	11.6	8.7	5.2	3.4
		MCWBR	4.2	4.1	4.8	7.6	6.3	5.7	5.7	5.6	6.4	6.1	5.1	3.4
Clear-Sky Solar Irradiance	taub	0.285	0.318	0.353	0.411	0.386	0.380	0.399	0.421	0.372	0.349	0.316	0.287	
	taud	2.096	2.059	2.040	2.129	2.328	2.378	2.327	2.252	2.371	2.345	2.241	2.088	
	Ebn at Noon	622	734	798	806	858	867	840	792	787	711	592	522	
	Edh at Noon	66	98	127	135	117	113	117	119	92	75	58	54	
All-Sky Solar Radiation	RadAvg	0.75	1.70	3.23	4.13	5.49	5.80	5.79	4.58	2.88	1.38	0.57	0.42	
	RadStd	0.11	0.21	0.31	0.43	0.38	0.53	0.54	0.43	0.41	0.23	0.10	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	-159	N/A	N/A

Nomenclature: See separate page